

MISMATHE MOCK TEST — 1

Class 11 · Maharashtra Board · MHT-CET Foundation · Combined PCM Test

Chapters: Chem — Some Basic Concepts | Phys — Units & Measurements | Maths — Trigonometry-I

Marks: 30 | Time: 40 min | Type: Single-correct MCQ

SECTION A — CHEMISTRY

Some Basic Concepts · Q1–Q10

Q1. The number of moles in 11 g of CO_2 (molar mass = 44 g/mol) is:

- (A) 0.25 (B) 0.5
(C) 1 (D) 0.025

Q2. The number of molecules present in 0.5 mole of O_2 is:

- (A) 6.022×10^{23} (B) 3.011×10^{23}
(C) 1.2×10^4 (D) 3.011×10^{22}

Q3. Carbon forms two oxides CO and CO_2 . For a fixed mass of carbon, the masses of oxygen are in ratio 1:2. This illustrates the law of:

- (A) Conservation of mass (B) Definite proportions
(C) Multiple proportions (D) Avogadro

Q4. The molarity of a solution containing 0.5 mol NaOH in 250 mL of solution is:

- (A) 1 M (B) 2 M
(C) 0.5 M (D) 4 M

Q5. The molar mass of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) is:

- (A) 160 g/mol (B) 172 g/mol
(C) 180 g/mol (D) 200 g/mol

Q6. The number of significant figures in 0.00340 is:

- (A) 2 (B) 3
(C) 4 (D) 5

Q7. "Equal volumes of all gases at the same temperature and pressure contain an equal number of molecules." This is:

- (A) Gay-Lussac's law (B) Avogadro's law
(C) Dalton's law (D) Boyle's law

Q8. The mass of 0.2 mole of water (H_2O , molar mass = 18 g/mol) is:

- (A) 1.8 g (B) 3.6 g
(C) 18 g (D) 9 g

Q9. A compound has empirical formula CH_2 and molecular mass 42. Its molecular formula is:

- (A) CH_2 (B) C_2H_4
(C) C_3H_6 (D) C_4H_8

Q10. The volume occupied by 1 mole of any gas at STP is:

- (A) 11.2 L (B) 22.4 L
(C) 44.8 L (D) 1 L

SECTION B — PHYSICS

Units & Measurements · Q11–Q20

Q11. The dimensional formula of force is:

- (A) $[\text{MLT}^{-1}]$ (B) $[\text{MLT}^{-2}]$
(C) $[\text{ML}^2\text{T}^{-2}]$ (D) $[\text{ML}^{-1}\text{T}^{-2}]$

Q12. Which of the following is a fundamental (base) SI unit?

- (A) newton (B) joule
(C) kelvin (D) watt

Q13. The number of significant figures in 6.320×10^3 is:

- (A) 2 (B) 3
(C) 4 (D) 5

Q14. The dimensional formula of work (or energy) is:

- (A) $[\text{MLT}^{-2}]$ (B) $[\text{ML}^2\text{T}^{-2}]$
(C) $[\text{ML}^2\text{T}^{-3}]$ (D) $[\text{ML}^{-1}\text{T}^{-2}]$

Q15. The percentage errors in measuring A and B are 2% and 3%. The maximum percentage error in the product $A \times B$ is:

- (A) 1%
- (B) 5%
- (C) 6%
- (D) 2.5%

Q16. Which of the following quantities is dimensionless?

- (A) velocity
- (B) strain
- (C) force
- (D) acceleration

Q17. The dimensional formula of pressure is:

- (A) $[ML^{-1}T^{-2}]$
- (B) $[MLT^{-2}]$
- (C) $[ML^2T^{-2}]$
- (D) $[ML^{-2}T^{-2}]$

Q18. Which of the following is NOT a use of dimensional analysis?

- (A) Checking correctness of an equation
- (B) Deriving a relation between quantities
- (C) Converting units between systems
- (D) Finding value of dimensionless constants

Q19. The number of significant figures in 0.00580 is:

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Q20. The "accuracy" of a measurement refers to:

- (A) closeness of repeated readings to each other
- (B) closeness of the measured value to the true value
- (C) the number of significant figures
- (D) the least count of the instrument

SECTION C — MATHEMATICS

Trigonometry-I · Q21–Q30

Q21. The value of $\sin 30^\circ + \cos 60^\circ$ is:

- (A) 1
- (B) $1/2$
- (C) $\sqrt{3}/2$
- (D) 0

Q22. In which quadrant are both $\sin \theta$ and $\cos \theta$ negative?

- (A) First
- (B) Second
- (C) Third
- (D) Fourth

Q23. $\sin(-\theta)$ is equal to:

- (A) $\sin \theta$
- (B) $-\sin \theta$
- (C) $\cos \theta$
- (D) $-\cos \theta$

Q24. The value of $\tan 45^\circ + \cot 45^\circ$ is:

- (A) 1
- (B) 2
- (C) 0
- (D) $\sqrt{2}$

Q25. If $\sin \theta = 3/5$ and θ is acute, then $\cos \theta$ equals:

- (A) $4/5$
- (B) $5/4$
- (C) $3/4$
- (D) $5/3$

Q26. $\sin(90^\circ + \theta)$ is equal to:

- (A) $\cos \theta$
- (B) $-\cos \theta$
- (C) $\sin \theta$
- (D) $-\sin \theta$

Q27. The value of $\cos 120^\circ$ is:

- (A) $1/2$
- (B) $-1/2$
- (C) $\sqrt{3}/2$
- (D) $-\sqrt{3}/2$

Q28. The period of the function $\tan \theta$ is:

- (A) 2π
- (B) π
- (C) $\pi/2$
- (D) 4π

Q29. $1 + \tan^2 \theta$ is equal to:

- (A) $\sec^2 \theta$
- (B) $\operatorname{cosec}^2 \theta$
- (C) $\cot^2 \theta$
- (D) 1

Q30. The value of $\sin 150^\circ$ is:

- (A) $1/2$
- (B) $-1/2$
- (C) $\sqrt{3}/2$
- (D) 1

ANSWER KEY & EXPLANATIONS

check only AFTER attempting

Q1	A	Q2	B	Q3	C	Q4	B	Q5	C
Q6	B	Q7	B	Q8	B	Q9	C	Q10	B
Q11	B	Q12	C	Q13	C	Q14	B	Q15	B
Q16	B	Q17	A	Q18	D	Q19	B	Q20	B
Q21	A	Q22	C	Q23	B	Q24	B	Q25	A
Q26	A	Q27	B	Q28	B	Q29	A	Q30	A

Chemistry

- **Q1 (A):** $n = \text{mass/molar mass} = 11/44 = 0.25 \text{ mol}$.
- **Q2 (B):** molecules = $0.5 \times 6.022 \times 10^{23} = 3.011 \times 10^{23}$.
- **Q3 (C):** fixed mass of one element, other in simple whole-number ratio = Law of Multiple Proportions.
- **Q4 (B):** $M = \text{mol/L} = 0.5/0.250 = 2 \text{ M}$.
- **Q5 (C):** $(12 \times 6) + (1 \times 12) + (16 \times 6) = 72 + 12 + 96 = 180 \text{ g/mol}$.
- **Q6 (B):** 0.00340 → leading zeros ignored; 3, 4, 0 → 3 sig figs.
- **Q7 (B):** definition of Avogadro's law.
- **Q8 (B):** mass = $n \times \text{molar mass} = 0.2 \times 18 = 3.6 \text{ g}$.
- **Q9 (C):** emp. mass = 14; $n = 42/14 = 3 \rightarrow \text{C}_3\text{H}_6$.
- **Q10 (B):** molar volume of any gas at STP = 22.4 L.

Physics

- **Q11 (B):** Force = mass $\times$ acceleration = $[\text{M}][\text{LT}^{-2}] = [\text{MLT}^{-2}]$.
- **Q12 (C):** kelvin is a base unit; newton/joule/watt are derived.
- **Q13 (C):** $6.320 \times 10^3 \rightarrow 6, 3, 2, 0 = 4 \text{ sig figs}$.
- **Q14 (B):** Work = force $\times$ distance = $[\text{MLT}^{-2}][\text{L}] = [\text{ML}^2\text{T}^{-2}]$.
- **Q15 (B):** for a product, % errors add: $2\% + 3\% = 5\%$.
- **Q16 (B):** strain = change in length/original length → dimensionless.
- **Q17 (A):** Pressure = force/area = $[\text{MLT}^{-2}]/[\text{L}^2] = [\text{ML}^{-1}\text{T}^{-2}]$.
- **Q18 (D):** dimensional analysis cannot find dimensionless constants (like 2, π).
- **Q19 (B):** 0.00580 → 5, 8, 0 = 3 sig figs.
- **Q20 (B):** accuracy = closeness to the true value (precision = closeness of repeats).

Mathematics

- **Q21 (A):** $\sin 30^\circ + \cos 60^\circ = \frac{1}{2} + \frac{1}{2} = 1$.
- **Q22 (C):** in Q3, sin – and cos – (only tan is +).
- **Q23 (B):** sin is an odd function → $\sin(-\theta) = -\sin\theta$.
- **Q24 (B):** $\tan 45^\circ + \cot 45^\circ = 1 + 1 = 2$.
- **Q25 (A):** $\cos\theta = \sqrt{1 - 9/25} = \sqrt{16/25} = 4/5$.
- **Q26 (A):** $90^\circ + \theta$ in Q2 (sin +); 90° changes function → $+\cos\theta$.
- **Q27 (B):** $\cos 120^\circ = \cos(180^\circ - 60^\circ) = -\cos 60^\circ = -\frac{1}{2}$.
- **Q28 (B):** period of tan is π .
- **Q29 (A):** Pythagorean identity: $1 + \tan^2\theta = \sec^2\theta$.
- **Q30 (A):** $\sin 150^\circ = \sin(180^\circ - 30^\circ) = +\sin 30^\circ = \frac{1}{2}$.

Scoring: +1 per correct, no negative marking. 25+/30 = excellent · 18–24 = solid · below 18 = revise weak spots & retake. | MISMATHE · Your MHT-CET Mentor